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Project Proposal Report on

Hierarchical Federated Learning for Privacy-Aware, Low-Latency Multimodal IoT Systems

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0.1 INTRODUCTION

The Internet of Things (IoT) has emerged as a transformative paradigm, enabling intelligent systems that leverage diverse sensors such as cameras, microphones, and environmental probes to enhance situational awareness across various domains. As these systems proliferate, they generate massive volumes of heterogeneous, multimodal data that must be processed to derive actionable insights [7]. Traditional approaches rely on transmitting raw data to centralized cloud servers for model training and inference. However, this centralized paradigm introduces significant challenges including high communication latency, excessive bandwidth consumption, and critical privacy vulnerabilities, as sensitive user data traverses untrusted networks [6, 12]. Therefore, there is a pressing need for distributed learning frameworks that can train collaborative AI models while preserving data locality and ensuring privacy, motivating the development of federated learning approaches tailored for edge–cloud IoT environments [9, 11].

The practical applications of privacy-preserving, low-latency multimodal learning span numerous critical domains. In Industrial IoT (IIoT), manufacturing systems can fuse camera feeds with vibration sensor data to enable predictive maintenance without exposing proprietary operational information. Smart city infrastructures benefit by combining traffic video streams with pollution sensor readings to optimize urban management while protecting citizen privacy. Healthcare systems can merge wearable device signals with medical imaging data to support real-time patient monitoring, where both latency and confidentiality are paramount. These applications benefit stakeholders including end-users, service providers, and governing bodies by enabling intelligent services that operate efficiently under strict privacy and resource constraints [2, 9].

Despite significant advances, existing federated learning frameworks exhibit critical limitations when applied to multimodal IoT scenarios [1]. Current privacy-centric federated learning approaches focus on gradient protection but operate in flat architectures that are unsuitable for high-bandwidth multimodal streams [5, 15]. Hierarchical frameworks like HED-FL optimize energy consumption but lack rigorous privacy mechanisms such as differential privacy [4]. Furthermore, cooperative approaches such as HCEF address device heterogeneity through compression techniques but do not guarantee strict privacy budgets or satisfy real-time inference latency requirements [14]. As a result, no existing solution comprehensively integrates hierarchical edge–cloud aggregation, formal privacy guarantees, and multimodal data fusion while meeting quantifiable Quality of Service (QoS) constraints [3, 8, 13].

This report proposes a novel hierarchical federated learning framework that ad-

dresses these limitations by enabling two-tier aggregation across devices, edge gateways, and cloud servers for multimodal data. The proposed system targets at least a 50% reduction in uplink communication overhead, integrates calibrated differential privacy with a strict privacy budget ($\varepsilon \leq 1$) while limiting accuracy degradation to within 2%, achieves inference latency below 100 milliseconds, and aims to achieve at least 20% energy savings per training round.

0.2 LITERATURE SURVEY

The existing literature on Federated Learning (FL) for IoT and edge computing can be broadly categorized into: (i) privacy-centric schemes, (ii) hierarchical and energy-efficient architectures, (iii) heterogeneity-aware cooperative learning, (iv) robust security frameworks, and (v) statistical optimization in hierarchical systems.

0.2.1 Privacy-Centric Schemes

Early privacy-preserving approaches relied on standard Differential Privacy (DP) mechanisms using Gaussian or Laplace noise, which often resulted in noticeable model accuracy degradation. Recent works such as FL2DP [5] have advanced this direction by combining DP with a shuffling mechanism. FL2DP employs an exponential mechanism to perturb gradients, thereby balancing data privacy with model utility, and introduces a shuffler entity to break the linkage between gradients and user identities (identity privacy). While effective for gradient protection, such flat architectures do not inherently optimize communication bandwidth for high-volume multimodal data streams.

0.2.2 Hierarchical and Energy-Efficient Architectures

To address the energy constraints of IoT devices, hierarchical federated learning architectures have been proposed. HED-FL [4] introduces a framework that brings intelligence closer to the edge using dynamic clustering of devices. By organizing devices into clusters, HED-FL reduces communication distance and energy consumption associated with long-range transmissions to a central server. Experimental results indicate that hierarchical clustering significantly extends network lifetime compared to flat FL approaches; however, the framework lacks rigorous privacy guarantees such as Differential Privacy.

0.2.3 Heterogeneity-Aware Cooperative Learning

System heterogeneity, arising from variations in device battery capacity, computational power, and network bandwidth, remains a critical bottleneck in federated learning. HCEF [14] proposes *Heterogeneity-Aware Cooperative Edge-based Federated Averaging* to mitigate the impact of straggler devices. The scheme employs an online control algorithm to dynamically adjust local update frequencies and gradient compression ratios based on device capabilities. This cooperative strategy ensures that faster devices

are not forced to idle while waiting for slower ones, thereby improving training efficiency, model accuracy, and energy consumption in mobile edge networks.

0.2.4 Robustness and Security Frameworks

Beyond privacy concerns, federated learning systems are vulnerable to poisoning and adversarial attacks. RoPPFL [15] introduces a hierarchical framework that simultaneously addresses privacy leakage and malicious model updates. The approach leverages edge nodes to perform similarity-based robust aggregation for detecting poisoned updates, while employing Local Differential Privacy (LDP) to protect user data. This dual-layer strategy demonstrates that edge nodes can effectively function as security verifiers without relying exclusively on centralized cloud-based validation.

0.2.5 Statistical Optimization in Hierarchical Systems

Performance degradation caused by non-IID (Independent and Identically Distributed) data is a major challenge in Fedderated Learning. Mhaisen *et al.* [8] addressed this by proposing a dynamic user-edge assignment framework within a two-layer federated architecture. By optimizing user assignment based on local data statistics, the framework creates statistically balanced edge-level datasets. The findings demonstrate that this approach significantly reduces model weight divergence and accelerates convergence compared to random assignment strategies.

0.2.6 Research Gaps

The analysis of existing literature reveals several unresolved research gaps. First, there is a lack of multimodal data fusion support, as most studies [4, 15] treat data as generic gradients and do not address the specific bandwidth challenges associated with fusing heterogeneous data types such as image and audio streams at the network edge. Second, a clear privacy–efficiency trade-off exists: HED-FL focuses on energy efficiency but does not incorporate Differential Privacy [4], whereas FL2DP enforces strong privacy guarantees without leveraging hierarchical bandwidth management mechanisms [5]. Third, real-time Quality of Service (QoS) constraints remain largely unaddressed [2], as existing robustness-oriented frameworks prioritize security verification over strict latency requirements (e.g., inference delays below 100 ms) necessary for real-time multimodal applications. Finally, there exists a conflict between privacy preservation and statistical optimization: although approaches such as that of Mhaisen *et al.* [8] improve convergence through statistical balancing, they rely on exposing local data distribution information (e.g., class labels), which violates the strict privacy budget ($\varepsilon \leq 1$) targeted by the proposed system, where performance optimization must be achieved without compromising user anonymity.

0.2.7 Recent Advances (2024–2025)

Recent work has validated the shift towards heterogeneity-aware and robust edge architectures. The HCEF framework (March 2025) demonstrated that adaptive compression is essential for mobile edge networks, successfully reducing training time by dynamically managing device capabilities. Similarly, FL2DP (April 2024) established that exponential noise mechanisms can achieve gradient indistinguishability with better accuracy than standard Gaussian methods.

Furthermore, RoPPFL (2024) confirmed that hierarchical edge computing is the optimal layer for deploying defense mechanisms against poisoning attacks. These findings directly support the proposed project's strategy of integrating Differential Privacy and Multimodal Fusion within a Hierarchical Edge-Cloud architecture.

0.3 PROBLEM STATEMENT

IoT systems increasingly combine diverse sensors (e.g. cameras, microphones, environmental probes) to improve situational awareness. However sending all raw multimodal data to the cloud incurs high latency and communication costs. On-device inference can reduce the volume of transmitted data and address privacy concerns by keeping sensitive processing local. We propose a novel hierarchical federated learning framework that jointly trains AI models across devices, edge gateways, and the cloud to satisfy strict QoS (latency, throughput, reliability) while enforcing privacy. The system will use two-tier aggregation (device $\rightarrow$ edge $\rightarrow$ cloud) to cut communication load and will inject calibrated differential-privacy noise in local updates to protect data. This approach is applicable to many domains: e.g. IIoT can fuse camera and vibration data, smart cities can combine traffic video and pollution readings, and healthcare can merge wearable and imaging data, all requiring real-time inference with strong privacy.

0.3.1 Objectives

The primary objectives of this work are:

- To design a hierarchical federated learning framework that operates in a two-tier architecture for multimodal data (e.g., image, audio, and sensor features), enabling efficient collaborative model training while significantly reducing communication overhead and latency compared to centralized learning approaches.
- To improve the efficiency and sustainability of federated learning systems by optimizing the trade-off between local computation and communication, targeting at least a 50% reduction in uplink traffic and achieving $\geq$20% energy savings per training round without compromising model performance.

- To integrate robust privacy-preserving mechanisms, such as differential privacy or secure aggregation, into local model updates, ensuring a strict privacy budget (e.g., $\varepsilon \leq 1$) while limiting the overall model accuracy degradation to $\leq 2\%$.
- To validate the proposed model through comprehensive simulation-based evaluation using realistic network and edge-computing environments (e.g., EdgeCloudSim and NS-3), with a focus on meeting Quality of Service (QoS) requirements such as low inference latency (< 100 ms) and high reliability ($> 99\%$ successful model updates) under varying network conditions.

0.4 PROPOSED METHODOLOGY

0.4.1 Flow Diagram

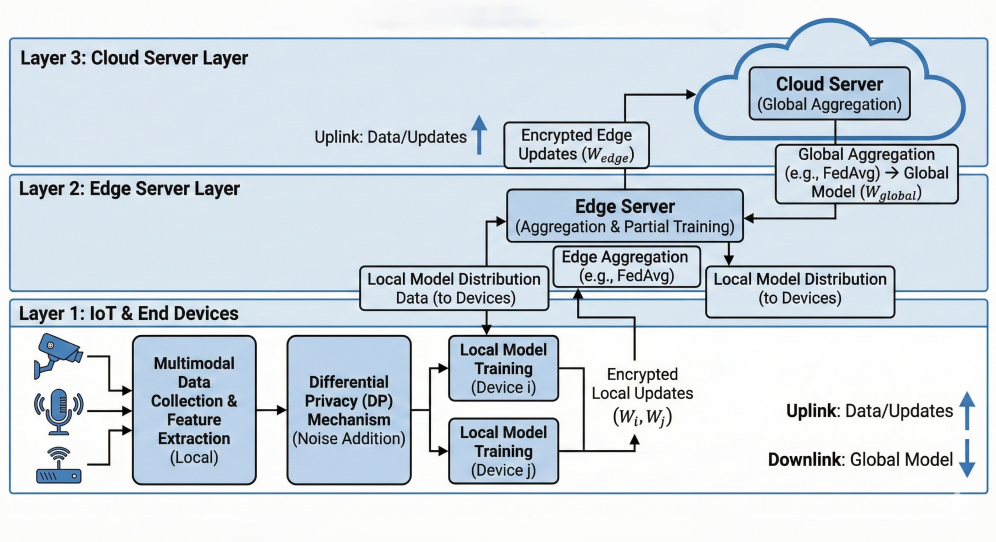


Figure 1: Process Flow of the Proposed System

0.4.2 Overall Pipeline

The proposed framework implements a Hierarchical Federated Learning (HFL) architecture operating across three layers: IoT/End Devices, Edge Servers, and a Cloud Server. Unlike traditional flat federated learning, the system adopts a two-tier aggregation strategy that balances local privacy preservation with global model accuracy. The pipeline begins with local multimodal data collection and feature extraction at end devices, followed by differential privacy (DP) noise injection during local training. Encrypted model updates are first aggregated at the edge layer to reduce communication overhead, after which a final global aggregation is performed at the cloud layer. This hierarchical design is engineered to meet strict Quality of Service (QoS) constraints, including inference latency below 100 ms and at least 20% energy savings per training round.

0.4.3 Multimodal Data Collection and Feature Extraction

IoT devices in the proposed system operate in heterogeneous environments and generate diverse data streams such as images, audio, vibration signals, and environmental readings. To mitigate bandwidth limitations associated with transmitting raw multimodal data, on-device feature extraction is employed. Raw sensor data is synchronized and locally processed, and lightweight feature representations are extracted instead of transmitting full-resolution data. This approach enables efficient multimodal fusion while significantly reducing communication cost before entering the privacy-preserving learning pipeline.

0.4.4 Stage 1: Privacy-Preserving Local Training

The first stage of training is executed entirely on end devices to ensure data locality and privacy. To enforce a strict privacy budget ($\varepsilon \leq 1$), a Differential Privacy mechanism is integrated into the local training loop. Calibrated noise (Gaussian or Laplace) is added to local gradients to achieve gradient indistinguishability. The local model is trained using on-device data, and only noisy, encrypted model updates are prepared for transmission. At no point does raw user data leave the device, ensuring strong protection for sensitive information.

0.4.5 Stage 2: Hierarchical Aggregation at Edge and Cloud

The core novelty of the proposed framework lies in its two-tier aggregation mechanism.

Edge-Level Aggregation

Edge servers act as intermediate aggregation nodes that collect encrypted updates from nearby devices. By dynamically clustering devices and performing partial aggregation at the edge, the system significantly reduces long-range communication to the cloud. This stage filters and compresses updates, targeting up to a 50% reduction in uplink traffic while lowering energy consumption on end devices.

Cloud-Level Aggregation

The cloud server performs the final global aggregation using the pre-aggregated updates received from edge servers. Since updates are already compressed and privacy-preserving, the cloud aggregation incurs lower computational overhead and reduced network latency, resulting in faster global model convergence.

0.4.6 QoS Validation and Optimization

To ensure real-time performance, the proposed methodology incorporates QoS validation through simulation-based evaluation using tools such as EdgeCloudSim or NS-3.

The hierarchical structure is designed to maintain inference latency below 100 ms, making it suitable for time-sensitive applications such as industrial monitoring and smart city services. Additionally, by balancing local computation with reduced communication frequency, the system targets energy savings of at least 20% per training round, extending the operational lifetime of battery-powered IoT devices.

0.4.7 Dataset

The proposed system is designed to support heterogeneous multimodal datasets representative of real-world edge–cloud environments. Evaluation is conducted using multimodal data relevant to application domains such as Industrial IoT (vibration and camera data), Smart Cities (traffic video and air quality sensors), and Healthcare (wearable sensor signals and medical imaging). Prior to training, data undergoes normalization and temporal synchronization to align different modalities. The processed data is then partitioned into non-IID subsets and distributed across edge devices to realistically simulate decentralized and heterogeneous IoT settings.

0.5 TIME LINE

Year	2026	2026	2026	2026	2026
Month	January	February	March	April	May
Literature Survey	★	★	★	★	★
Problem Formulation		★			
Objective – I (Framework Design)			★		
Objective – II (Optimization & Efficiency)				★	
Objective – III (Privacy Integration)					★
Simulation & Evaluation				★	★
Paper Submission					★
Report Submission					★

Table 1: Proposed timeline for the project (January–May 2026)

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